

SAMBIT BEHURA

AI / GenAI Engineer · Data Scientist

Bhubaneswar, Odisha, India | sambitkx2020@email.iimcal.ac.in | [linkedin.com/in/behura-sambit](https://www.linkedin.com/in/behura-sambit) | github.com/sbehu

SUMMARY

AI/GenAI engineer and data scientist with 9+ years of professional experience — including 6.5+ years across McKinsey, American Express, and consumer lending — now focused on productionizing agentic AI systems. Built and deployed multi-agent RAG pipelines (CrewAI, LangGraph, LangChain, Pinecone/ChromaDB), exposed tools via a custom MCP server, and built a multimodal vision-LLM document-verification agent (GPT-4o Vision, AWS Textract, AWS Rekognition) on AWS ECS Fargate with CI/CD and rigorous evaluation frameworks, on top of a core background in ML-driven decisioning — XGBoost/logistic-regression scorecards, credit risk analytics, and causal experimentation that quantified ~\$200M in incremental business impact.

EDUCATION

Post Graduate Diploma in Business Analytics (PGDBA) — IIM Calcutta, ISI Kolkata & IIT Kharagpur (2018 – 2020)
B.Tech, Electrical and Electronics Engineering — National Institute of Technology, Tiruchirappalli (2011 – 2015)

SKILLS

GenAI / LLM: Multi-agent orchestration (CrewAI, LangGraph), LangChain, RAG architecture, Pinecone, ChromaDB, hybrid retrieval (dense + BM25 + cross-encoder rerank), vision-language models (GPT-4o Vision), prompt engineering, OpenAI Structured Outputs (Pydantic), LLM-as-judge evaluation (RAG Triad), Model Context Protocol (MCP)

ML / Data Science: Python, XGBoost, LightGBM, Random Forest, Logistic Regression, SHAP, time series analysis, hypothesis testing, test-control experimentation, GINI/PSI/CSI model monitoring, credit risk & scorecard modeling

MLOps / Cloud: AWS (ECS Fargate, ECR, S3, CloudWatch, Rekognition, Textract), Docker, GitHub Actions CI/CD, Streamlit, SQL

Certifications: Building and Evaluating Advanced RAG · LangChain for LLM Application Development

INDEPENDENT GENAI PROJECTS

Chargeback Intelligence Agent — *Compliance-grade RAG pipeline for card-network dispute resolution*
github.com/sbehu/chargeback-intel-agent

- Built a deterministic, auditable RAG pipeline (Pinecone + BM25 + rerank, OpenAI Structured Outputs via Pydantic) grounding chargeback defenses in real Visa and Mastercard rule text, validated by an LLM-as-judge RAG Triad harness (400 CFPB cases, 0.97 answer relevance) with full audit logging to S3; deployed on AWS ECS Fargate via GitHub Actions CI/CD.

AI Financial Analyst Agent — *Multi-agent RAG system for 10-K financial analysis*
github.com/sbehu/financial_agent_intel

- Orchestrated a CrewAI + LangGraph workflow to parse 6 years of American Express 10-K filings and benchmark them against live, web-sourced competitor context, pairing a self-correcting LangGraph retrieval pipeline (LLM-judged relevance retry) with a math-verification agent.
- Exposed core tools via a custom MCP server for cross-agent reuse, and added a CrewAI report-generation stage; evaluated against a 200+ question golden set and deployed via Docker on AWS ECS Fargate.

Loan Document Fraud Detection Agent — *Multimodal vision-LLM verification pipeline for consumer-lending underwriting*
github.com/sbehu/loan-doc-verifier

- Built an 8-check GPT-4o vision + AWS Textract and Rekognition pipeline verifying signatures, balance reconciliation, identity, income, and cross-applicant document recycling, backed by an agentic investigation step and a deterministic, auditable verdict engine; stress-tested against real documents and deployed on AWS ECS Fargate via GitHub Actions CI/CD.

EXPERIENCE

Senior Consultant — EXL

Mar 2026 – Apr 2026

- Engineered features for an existing fraud-detection model on a Long Term Care (LTC) insurance client engagement, using claims behavioral data to help flag anomalous claim patterns.

Assistant Manager, Global Commercial Services — American Express

Jun 2022 – Nov 2025

- Built an XGBoost-based B2B product recommendation model (GINI 0.85) to identify optimal buyer-supplier matches from accounts-payable data, engineering 70+ features with cross-validation and hyperparameter tuning, and improved stakeholder trust via SHAP-based explainability.
- Quantified ~\$200M in annual incremental billed business by measuring marketing campaign impact through digit-based test-control experimentation and bootstrapped confidence intervals, directly informing leadership's campaign investment decisions.

Senior Manager, Analytics — SMFG India Credit (Fullerton India)

Dec 2021 – Jun 2022

- Monitored and validated Application, Behavioural, and Collection credit risk models using GINI, PSI, and CSI stability metrics, flagging drift and recommending recalibration.

Data Scientist — U GRO Capital

Apr 2020 – Nov 2021

- Built U GRO's first-generation business-loan application scorecard, engineering 1,000+ features from bank statement and bureau data and shipping a logistic regression model (GINI ~0.80) that became the core of credit underwriting.

Data Science Intern (Pre-Placement Offer) — McKinsey & Company

Oct 2019 – Mar 2020

- Developed a personal-loan cross-sell model for early-tenure consumer-durable and two-wheeler customers using XGBoost and Random Forest.

Early Career (~3 yrs): Asst. Professor of Mathematics, FIITJEE, Bhubaneswar (2016–18) · Business Analyst, RBS, Chennai (2015)